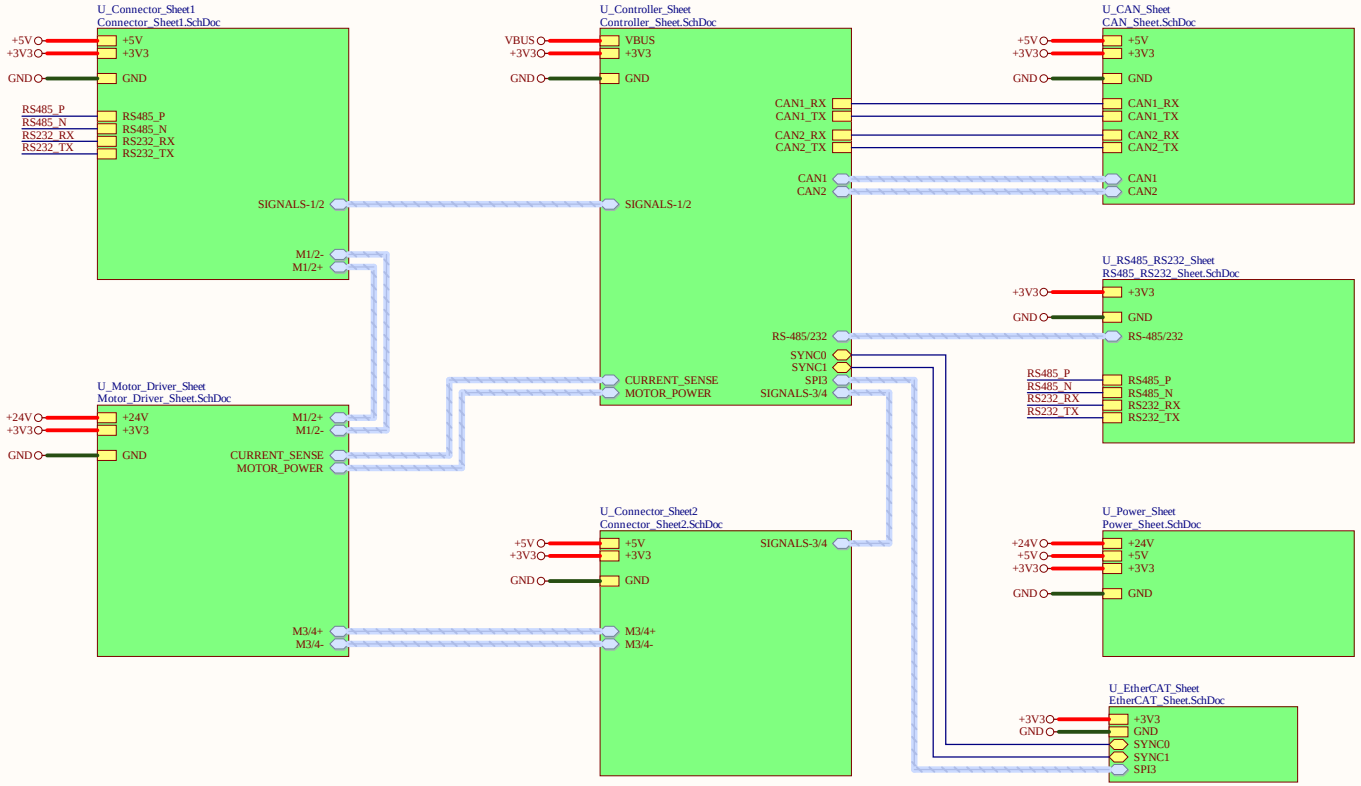
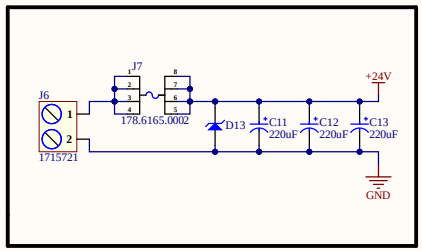
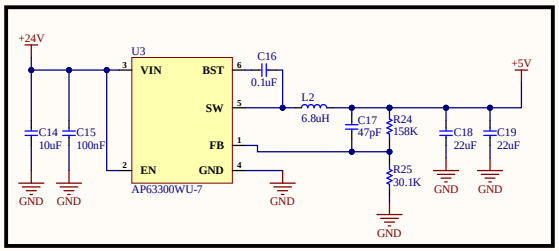


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Date: 30-Aug-24	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 9

24V INPUT

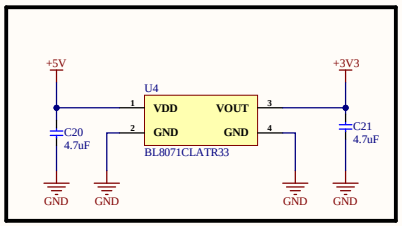


5V 3A BUCK



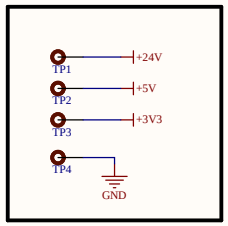
3A

3.3V 1.5A BUCK

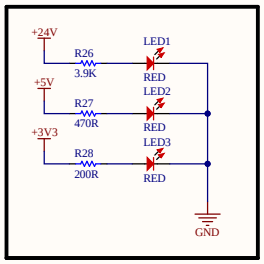


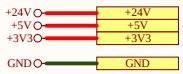
1.5A

TEST POINTS



LEDs

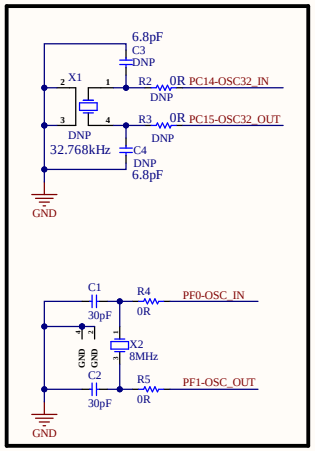




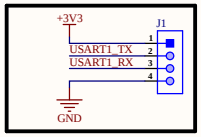
eSpark Consulting Group

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Date: 30-Aug-24	Page Num: 2
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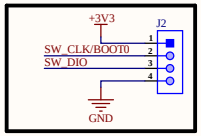
CRYSTAL



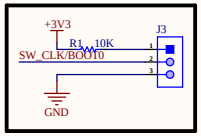
DEBUG

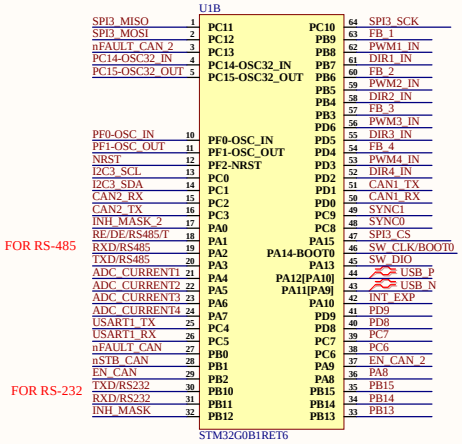


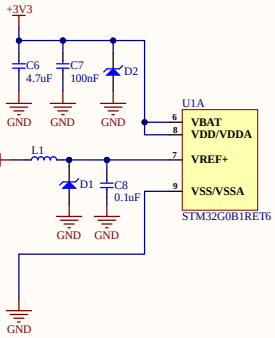
SWD PORT



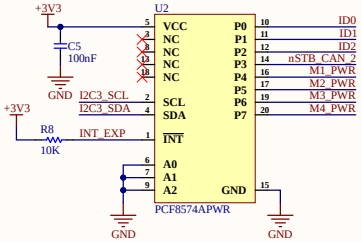
BOOT



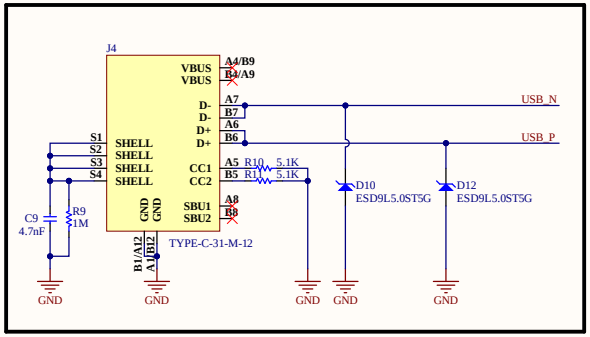




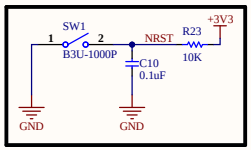




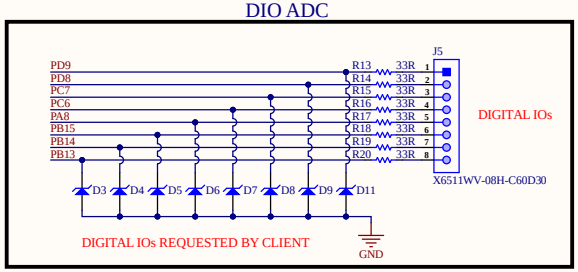
USB

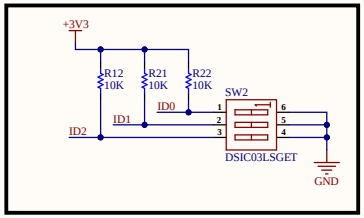


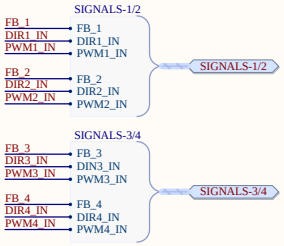
RESET

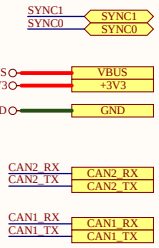


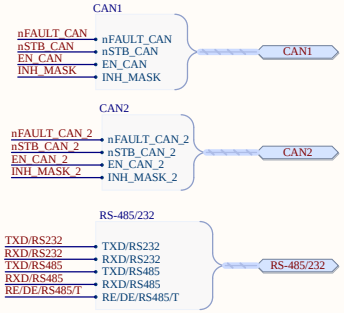
DIO ADC

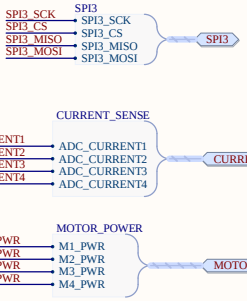








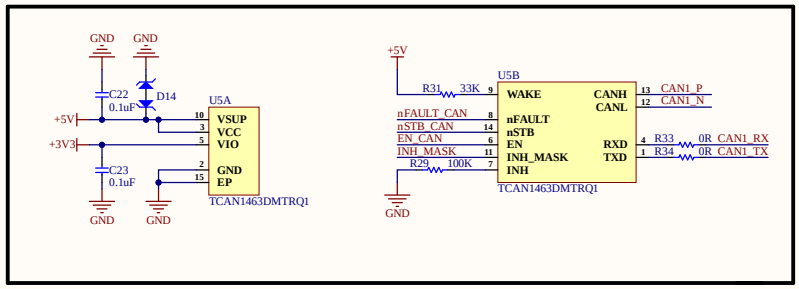




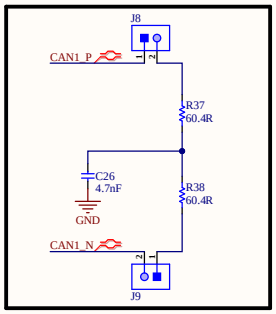
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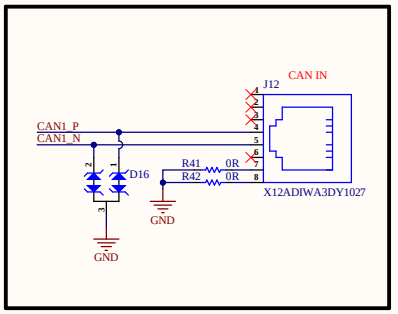
CAN BUS



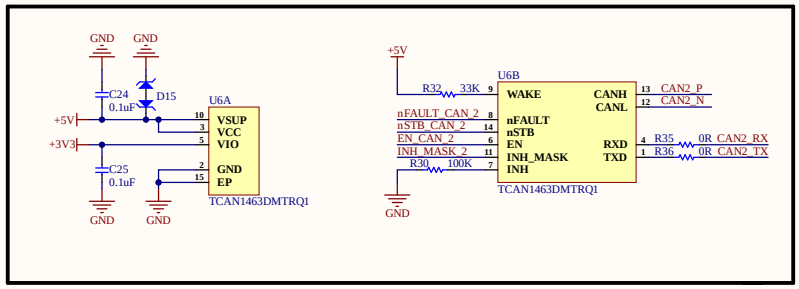
CAN TERMINATION



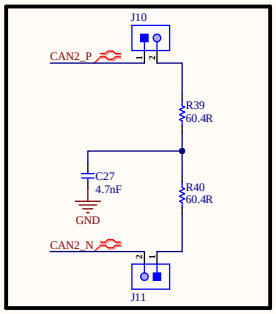
CAN CONNECTOR



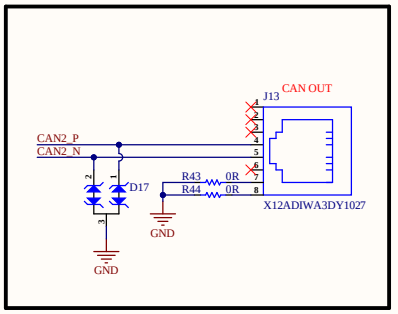
CAN BUS

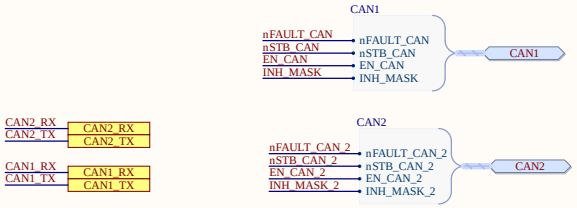


CAN2 TERMINATION



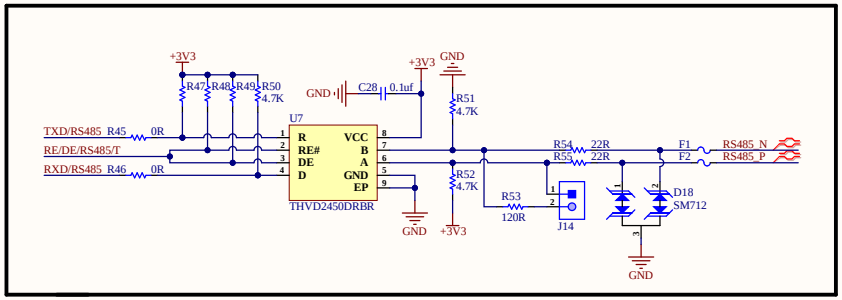
CAN2 CONNECTOR



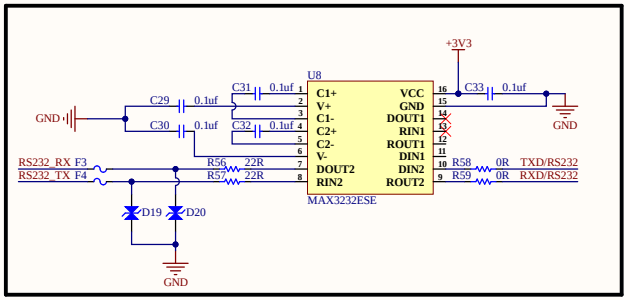


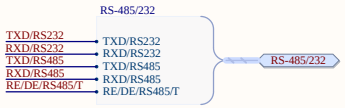
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Date: 30-Aug-24	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 9

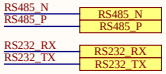
RS485 TO UART



RS232 TO UART





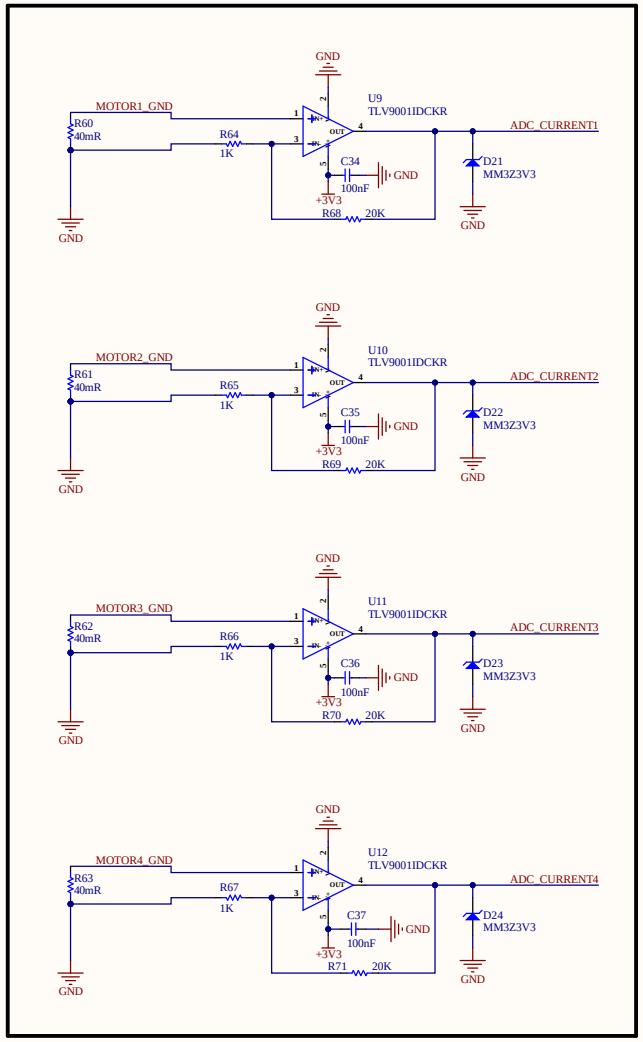




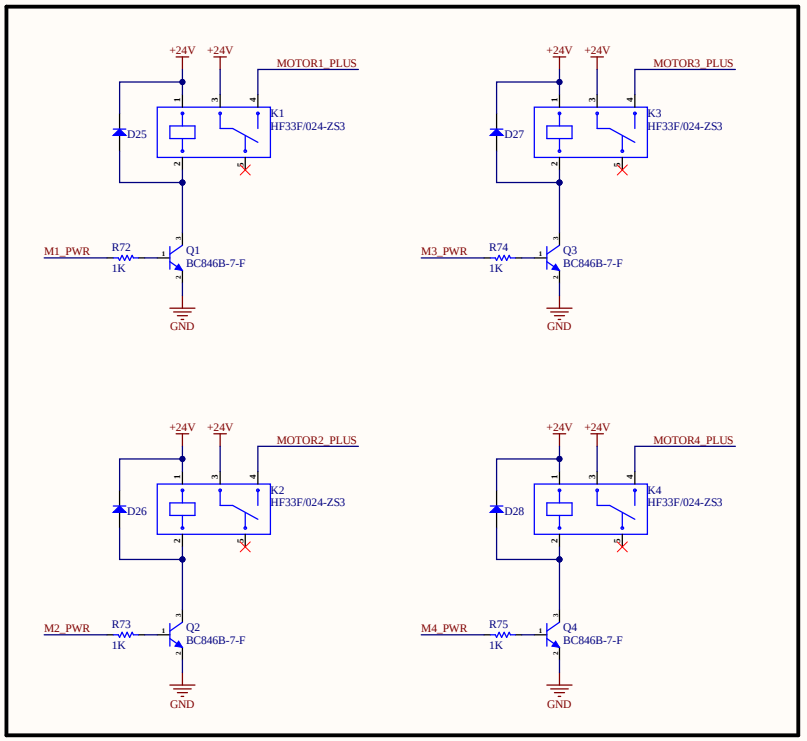
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File: BLDC-Board-v1.0.PrjPcb	
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Drawn By: Huzaifa Ali	Page Total: 9

CURRENT SENSE



MOTOR POWER REALY



Vout = (1+ 13.7k/1k) * 0.16v(max)
@4A
Vout = 3.19V

MOTOR_POWER
M1_PWR → M1_PWR
M2_PWR → M2_PWR
M3_PWR → M3_PWR
M4_PWR → M4_PWR
→ MOTOR_POWER

M1/2-
MOTOR1_GND → MOTOR1_GND
MOTOR2_GND → MOTOR2_GND
→ M1/2-

M1/2+
MOTOR1_PLUS → MOTOR1_PLUS
MOTOR2_PLUS → MOTOR2_PLUS
→ M1/2+

M3/4-
MOTOR3_GND → MOTOR3_GND
MOTOR4_GND → MOTOR4_GND
→ M3/4-

M3/4+
MOTOR3_PLUS → MOTOR3_PLUS
MOTOR4_PLUS → MOTOR4_PLUS
→ M3/4+

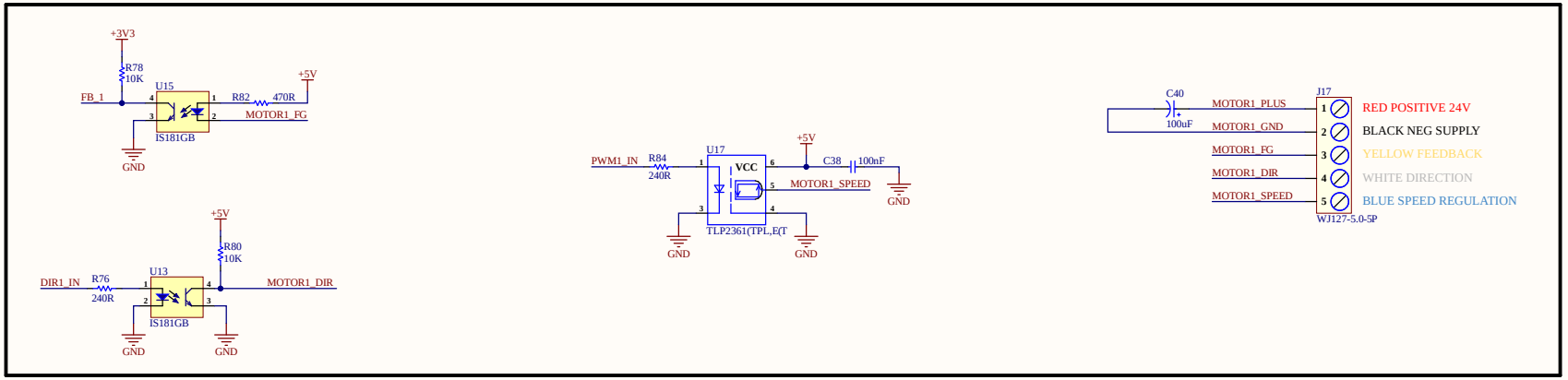
CURRENT_SENSE
ADC_CURRENT1 → ADC_CURRENT1
ADC_CURRENT2 → ADC_CURRENT2
ADC_CURRENT3 → ADC_CURRENT3
ADC_CURRENT4 → ADC_CURRENT4
→ CURRENT_SENSE

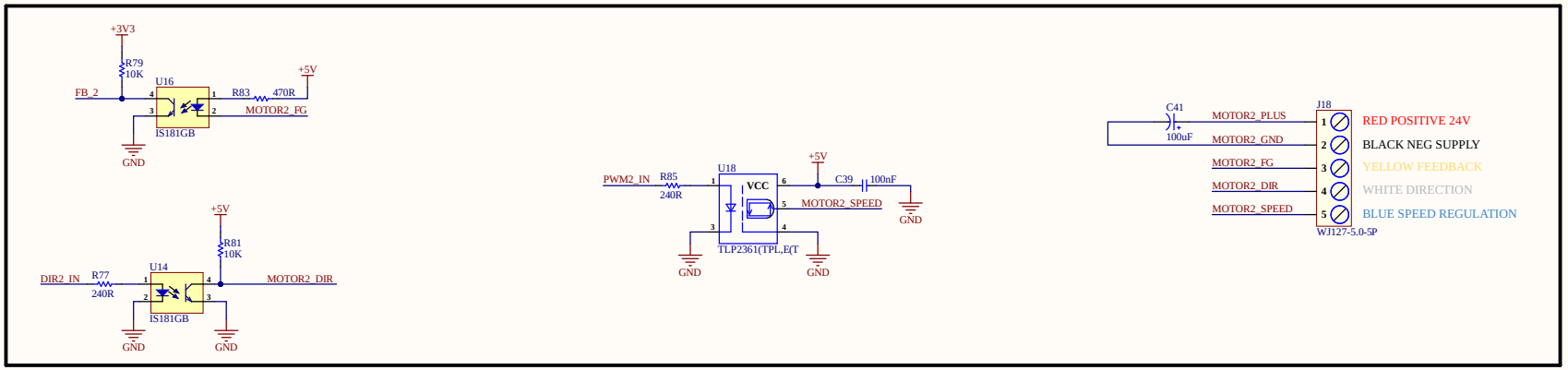
+24V → +24V
+3V3 → +3V3
GND → GND

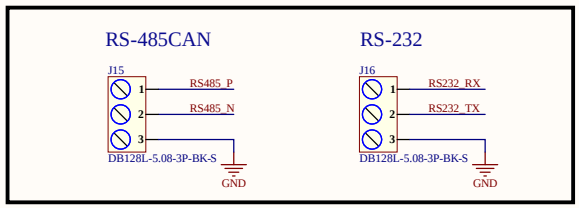
eSpark Consulting Group

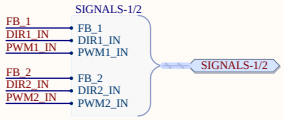
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File: BLDC-Board-v1.0.PrjPcb	
Date: 30-Aug-24	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 9

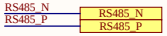
MOTOR SIGNALS

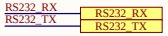


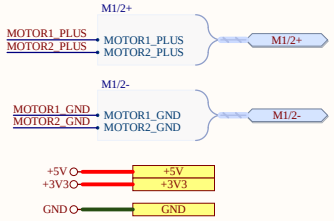








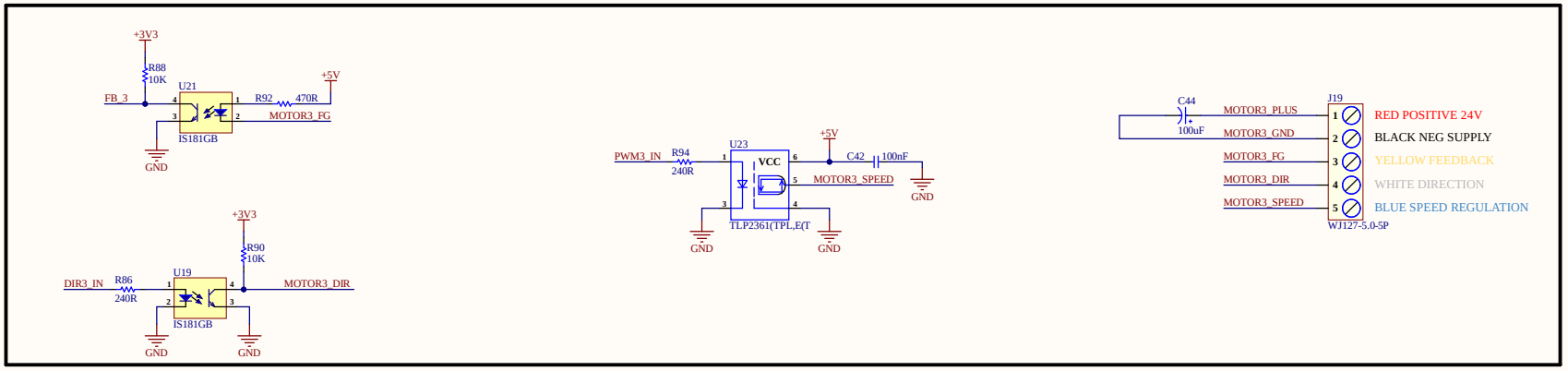


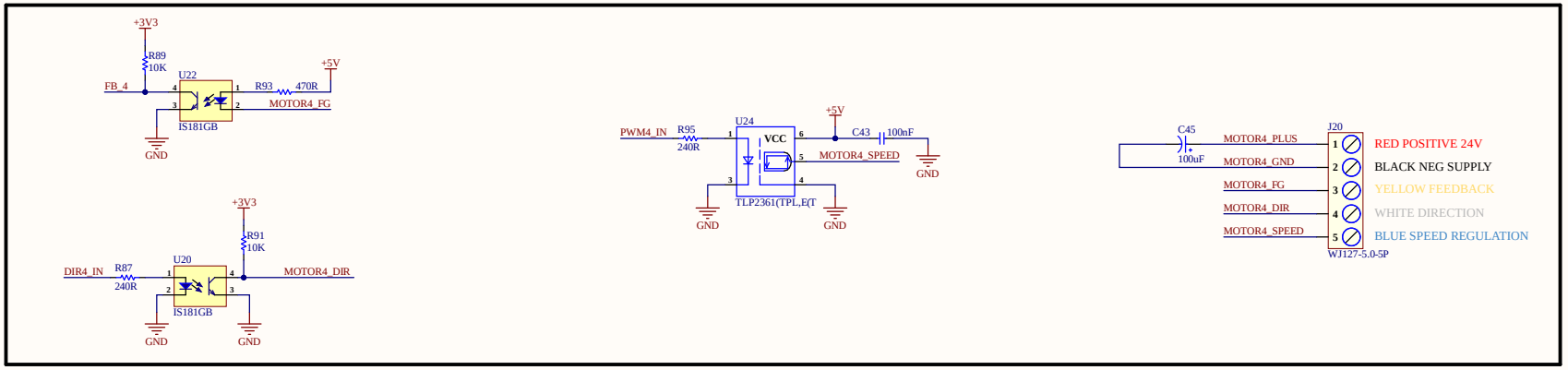


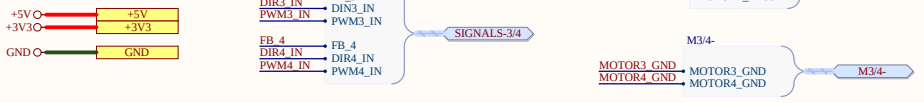
eSpark Consulting Group

Title: Connector_Sheet1.SchDoc	
File: BLDC-Board-v1.0.PrjPcb	
Date: 30-Aug-24	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 9

MOTOR SIGNALS

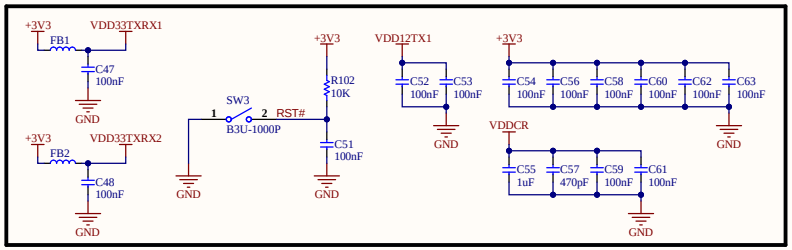




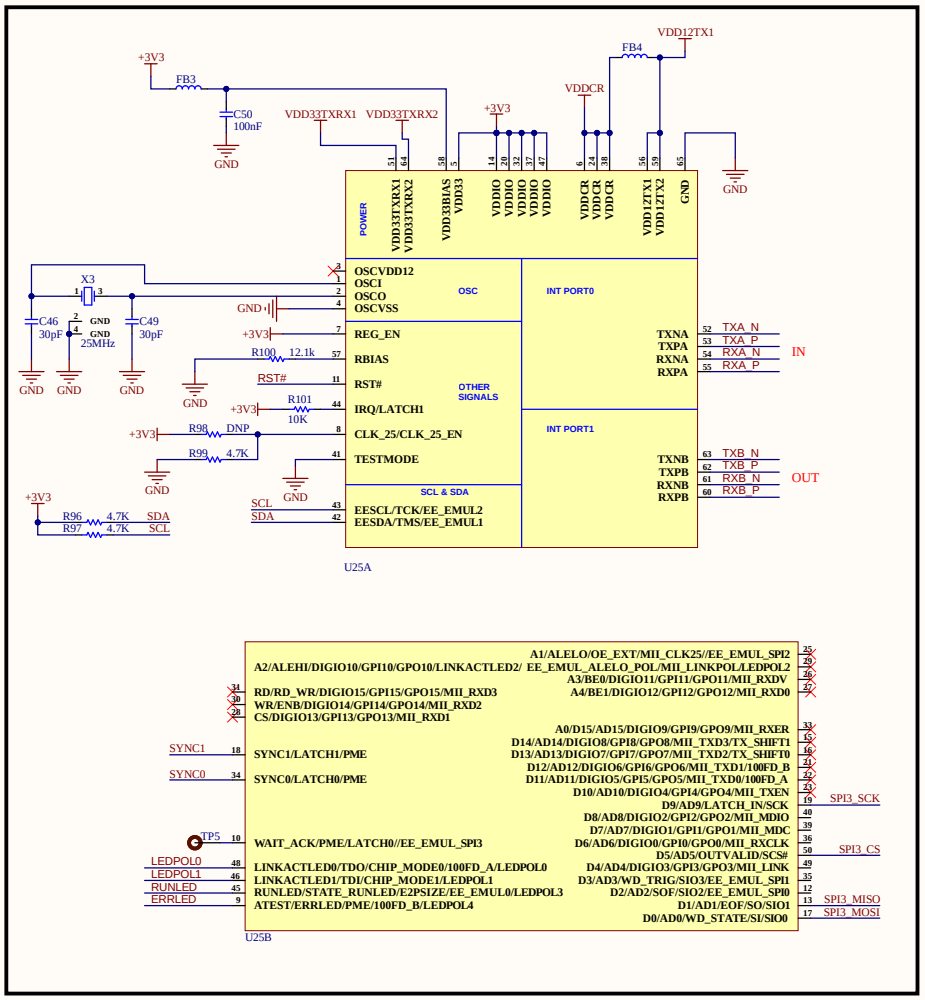


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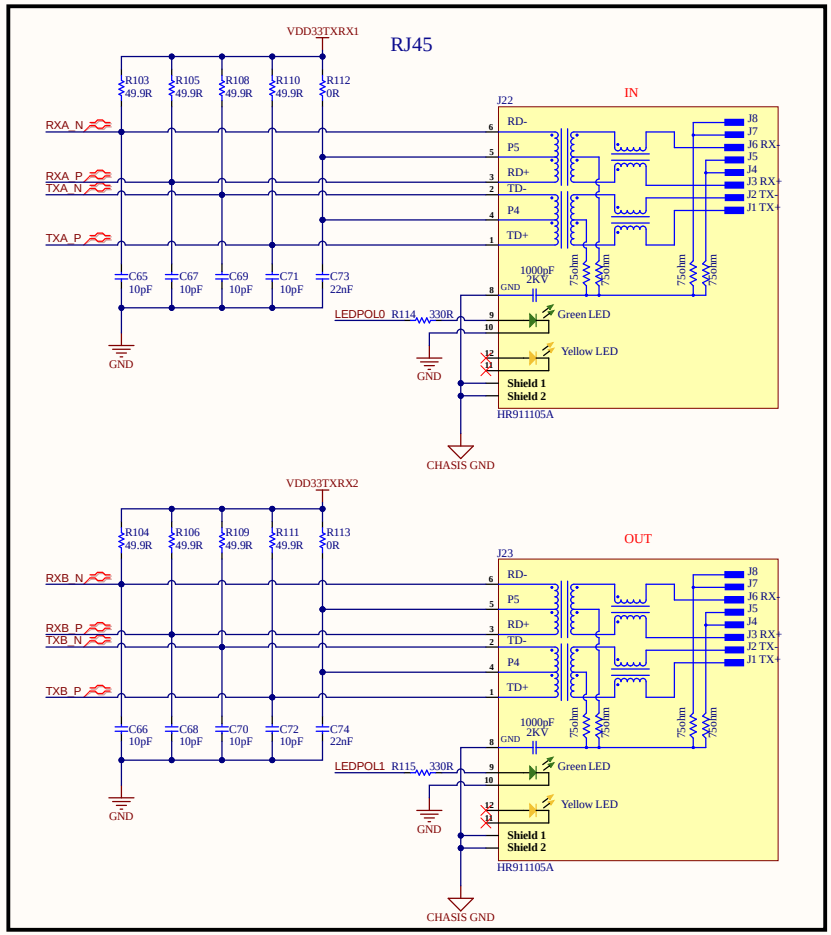
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File: BLDC-Board-v1.0.PrjPcb	
Date: 30-Aug-24	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 9

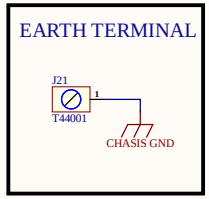


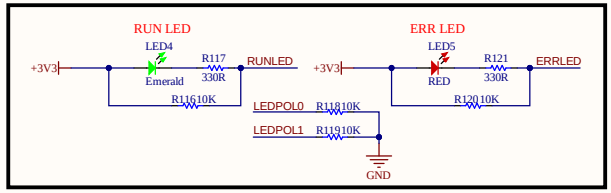
ETHERCAT

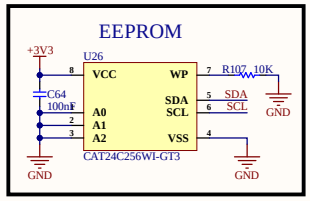


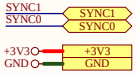












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Title: EtherCAT_Sheet.SchDoc		
File: BLDC-Board-v1.0.PrjPcb		
Date: 30-Aug-24	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 9